

Solutions – Quiz 6

ECON 320 – Introduction to Mathematical Economics

Instructor: Muhebullah Karimzada

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Part A — Multiple Choice

1. (c)
2. (b)
3. (a)
4. (b)
5. (c)
6. (b)
7. (d)

Part B — Short Answer Solutions

Question 1 — Envelope Theorem

(1) Derivative of the value function

From the Envelope Theorem for a constrained maximization problem:

$$\frac{dV}{da} = U_a + \lambda g_a$$

evaluated at the optimum.

(2) Why indirect effects disappear

The FOCs ensure that the derivatives with respect to x and y cancel out. Thus changes in a affect $V(a)$ only directly through U_a and indirectly through the constraint term g_a .

(3) Interpretation of $-\lambda g_a$ (or $+\lambda g_a$ depending on constraint form)

It measures how changes in the parameter a affect the tightness of the constraint, multiplied by the shadow value of relaxing the constraint. It is the effect on the optimized value caused by changes in the feasible set.

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Question 2 — Duality and Shephard's Lemma

(1) Conditional factor demands

By Shephard's Lemma:

$$\frac{\partial C}{\partial w} = L^*, \quad \frac{\partial C}{\partial r} = K^*.$$

(2) Duality explanation

Cost minimization and output maximization share the same tangency condition ($MRTS = w/r$). Therefore the cost function and production function describe the same technology from two perspectives: one minimizes cost for a given output, the other maximizes output for given inputs.

(3) Marginal cost

$$MC(Q) = \frac{\partial C}{\partial Q}$$

because the Lagrange multiplier in the cost-minimization problem equals marginal cost.

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Question 3 — Capital Dynamics

The differential equation is:

$$\frac{dK}{dt} = I(t) - \delta K.$$

Solution:

$$K(t) = K(0)e^{-\delta t} + \int_0^t e^{-\delta(t-s)} I(s) ds.$$

Steady state (constant I):

$$K^* = \frac{I}{\delta}.$$

Interpretation: - $K(0)e^{-\delta t}$ = initial capital depreciated over time. - The integral term = accumulated past investment adjusted for depreciation.

Question 4 — Present Value

Income:

$$Y(t) = Y_0 e^{gt}.$$

Present value:

$$PV = \int_0^\infty Y_0 e^{(g-r)t} dt = \frac{Y_0}{r-g}, \quad r > g.$$

Condition for finiteness:

$$r > g$$

because income must not grow faster than it is discounted.